

Project Development Phase

Model Performance Test — Cyber Security with AI (VAPT)

Date: 29 June 2026

Team ID: N/A

Project Name: Network Anomaly Detection

Maximum Marks: 10 Marks

Model Performance Testing:

S.No.	Parameter	Values	Screenshot
1.	Information Gathering Footprinting — Reconnaissance —	NOT APPLICABLE Footprinting and reconnaissance are VAPT phases that involve actively probing a target system or network to collect information prior to a penetration test. This project is a detection system, not a penetration-testing engagement. No target was probed.	N/A
2.	Scanning the Target Scanning info — Risk factors —	NOT APPLICABLE Network/port scanning and risk-factor assessment are offensive VAPT activities. This project performs anomaly detection on existing historical network flow data; it does not scan any live target.	N/A
3.	Gaining Access Access process — Vulnerability found —	NOT APPLICABLE Gaining access is an exploitation phase of VAPT. This project classifies network flows as Normal or Anomaly using Isolation Forest; it does not exploit vulnerabilities.	N/A
4.	Maintaining Access Automation (AI implementation) AI tools used — Automation implemented —	NOT APPLICABLE — Maintaining Access Maintaining access (persistence) is a post-exploitation VAPT phase outside this project's scope. Closest mapping for AI / Automation: Algorithm: Isolation Forest (scikit-learn) — unsupervised anomaly detection on 2.52 M network flows.	N/A

S.No.	Parameter	Values	Screenshot
		AI tools: Python 3, scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, joblib — run in Google Colaboratory. Automation: Full pipeline (data cleaning → feature scaling → model training → prediction → dashboard generation) automated in a single Jupyter notebook; trained model serialised with joblib for reuse without retraining. This information is documented in detail in the ML template.	
5.	Covering Tracks & Report Vulnerability risk factors — VAPT report —	NOT APPLICABLE Covering tracks is the final VAPT phase in which evidence of the penetration test is removed. This project does not conduct VAPT. Risk factors identified by the anomaly detector are discussed in the project manual (Milestones 5, 7, and 11) as proposed future work for production incident-response integration, not as active VAPT findings.	N/A

Note: This template is designed for Vulnerability Assessment and Penetration Testing (VAPT) engagements — i.e. projects that actively probe, exploit, and report on vulnerabilities in a target system. The Network Anomaly Detection project is a defensive AI/ML tool that detects anomalous network traffic patterns; it is not an offensive VAPT exercise. Real-time deployment and incident-response automation are documented as design proposals (Milestones 6 and 11 of the project manual) rather than implemented VAPT activity.